3rd CII National Kaizen Circle Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

24th ~ 26th June 2020

Company Profile

COMPANY OVERVIEW

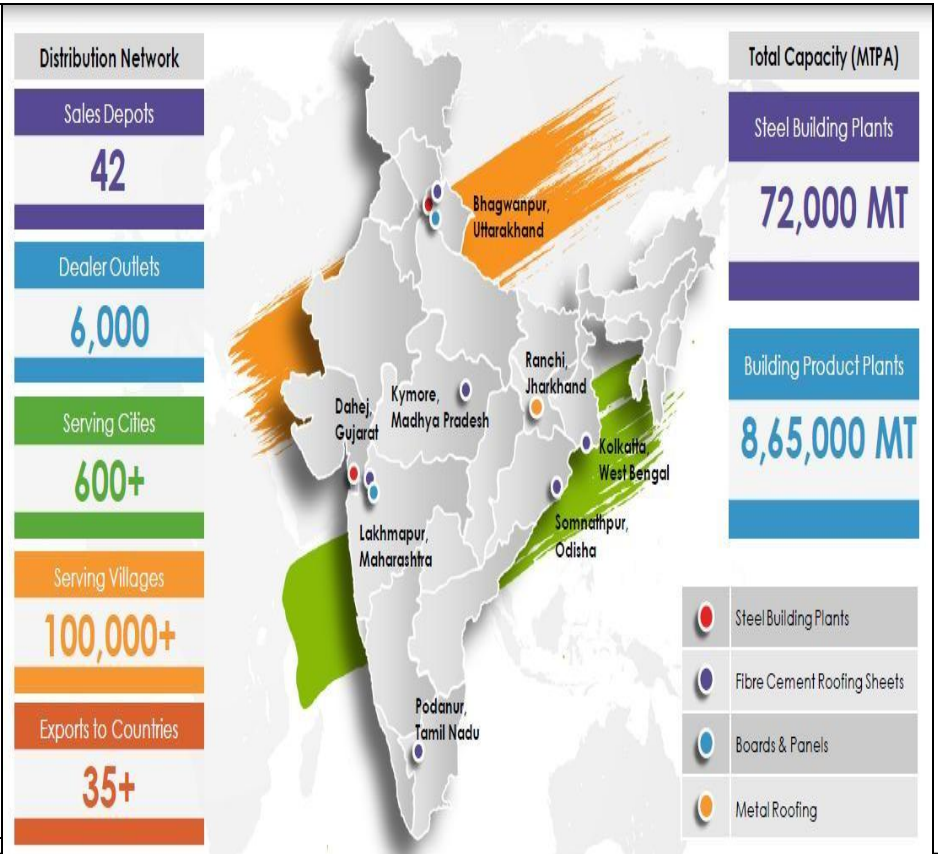
Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

OUR BUSINESS

The company businesses are divided into two distinct business segments
with capacity –
1.Buildings Products – 8,65,000 MT / Annum
2.Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company's revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Company Product & Customers

Products:


Fibre Cement Roofing Sheets


Hi-Tech Roofing Sheets


Super Sheets


Cement Planks


Fibre Cement Boards


Rapicon Walls


Smart Steel Buildings

Customers:

Reliance Industries Limited
Simplex Infrastructures Limited
Walmart Save money. Live better.
IFB
ACC
Mahindra
HYATT CAPITAL GATE ABU DHABI
Pidilite
TATA
Whirlpool CORPORATION
RELIANCE
Marriott HOTELS & RESORTS
ITC Limited
BRITANNIA Eat Healthy, Think Better
DLF
DLF BUILDING INDIA
एन बी सी सी NBCC A Navratna CPSE
Cipla
Coca-Cola
VIDEOCON

Kaizen Theme

Kaizen Theme: To eliminate Machine stoppages for Cutters & Shoes change.

Implemented Area: Line No -1, Podanur Works, Coimbatore

Implementation Date: 04th November 2019

Cross Functional Team:

Mr. M. Palavesa Moorthy
(Junior Engineer Production)

Team Leader
Mr. Y. Kesavaraj
(Senior Engineer Production)

Mr. N. Anil Kumar Reddy
(Engineer - QS & BE)

Mr. M. Madhan Raj
(Shift Incharge Production)

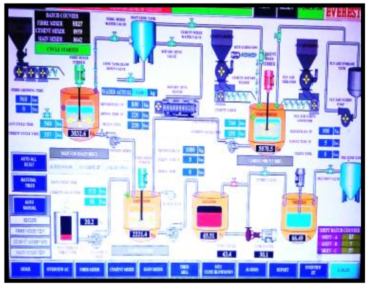
Mr. N. Karthikeyan
(Engineer Production)

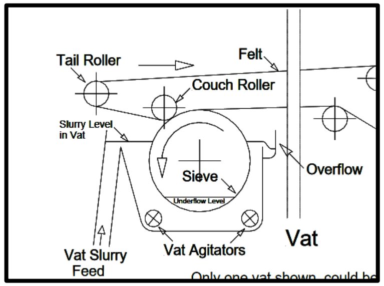
Mr. G. Tulasi Rao
(Shift Incharge - Production)

Mr. Jahir Hussain
(Fitter - Maintenance)

Problem / Present Status

Process Flow:



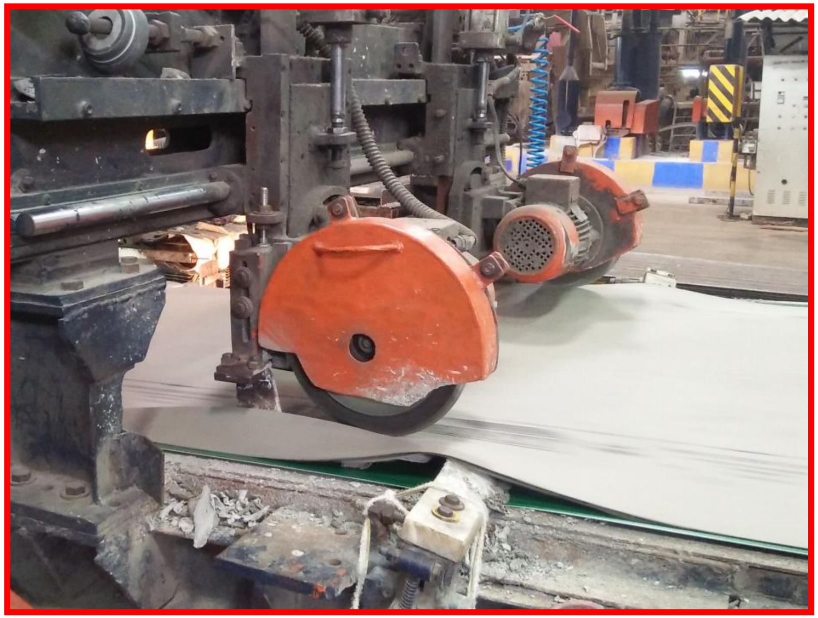


1) RAW MATERIAL PREPARATION → 2) VATS → 3) SHEET FORMING → 4) SHEET CUTTING → 5) SHEET HANDLING / PILING MACHINE → 6) PRE CURING → 7) DE-PILING

5) SHEET HANDLING / PILING MACHINE → SHEET RECYCLING UNIT

Problem / Present Status

TCT Cutter



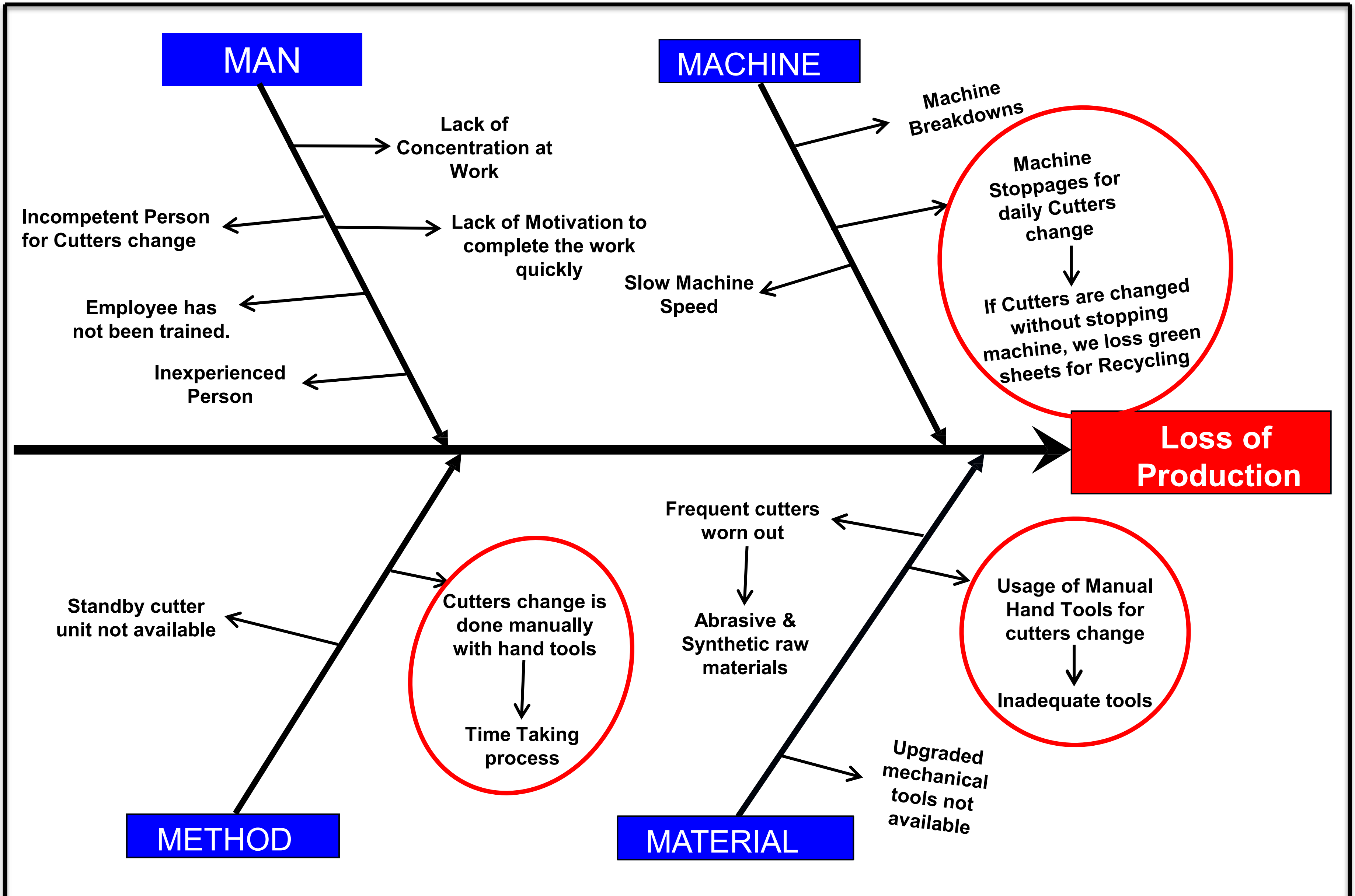
TCT Cutter Unit

Problem/ Present Status :

Loss of Production i.e., 3.54 MT/Day production due to Machine Stoppage for 15 minutes daily for cutters change.

- Due to abrasive raw materials, while cutting TCT cutters sharpness get blunt. Due to this sheet cutting finishing will not be good. For this we have to change TCT cutters in every 24 Hours.

Root Cause Analysis



Root Cause - Why Why Analysis

Problem Statement	Loss of Production	
Why 1	Due to Machine stoppages.	Time taking process for changing cutters & shoes.
Why 2	Machine stoppages for Cutters & Shoes Change.	Cutters & shoes are changed manually by using hand spanners.
Why 3	If Machine not stopped, we loss wet green sheets for recycling.	There is no usage of efficient tool to change cutters & shoes quickly.
Why 4	There is NO process to collect Green wet sheets while changing cutters & shoes to eliminate sheet wastage.	

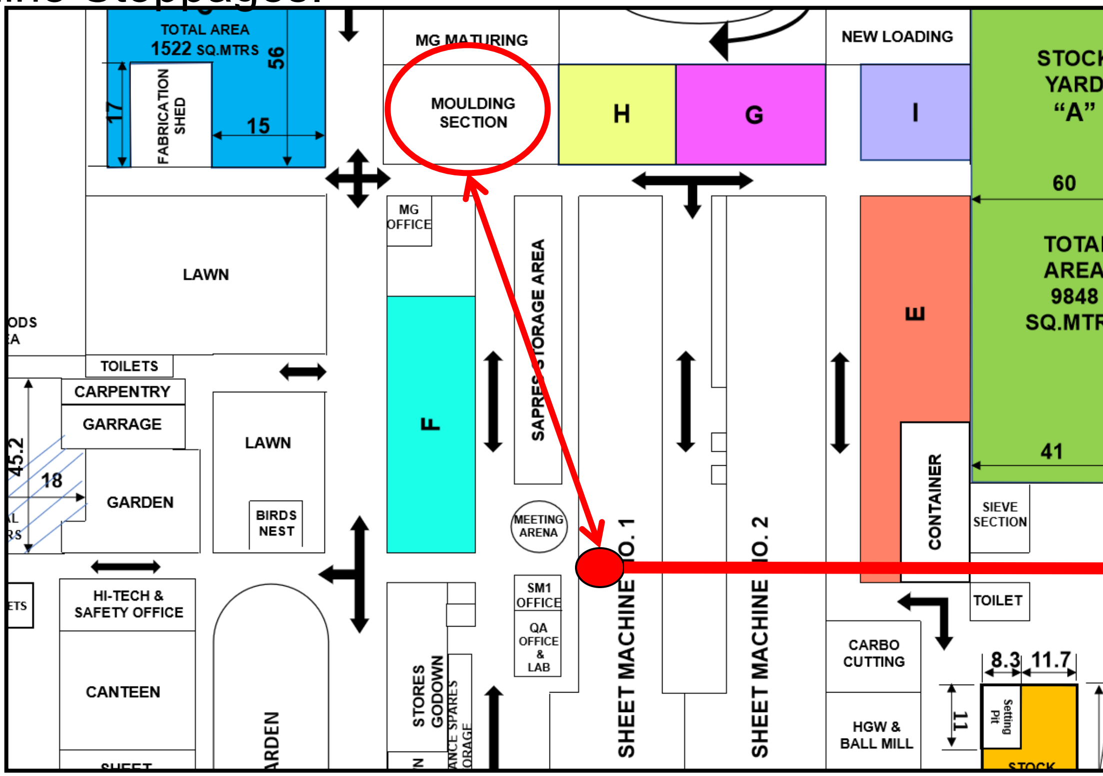
Root Cause:

1) There is NO Process to collect Green sheets during Cutters change.

2) There is NO usage of efficient tool to change cutters & shoes quickly.

Idea Generation

<u>Idea Generation: 1.</u> To change Cutters & Shoes while giving sheets for molding.
2. To use Pneumatic wrench to quickly change cutters to eliminate Machine Stoppages.



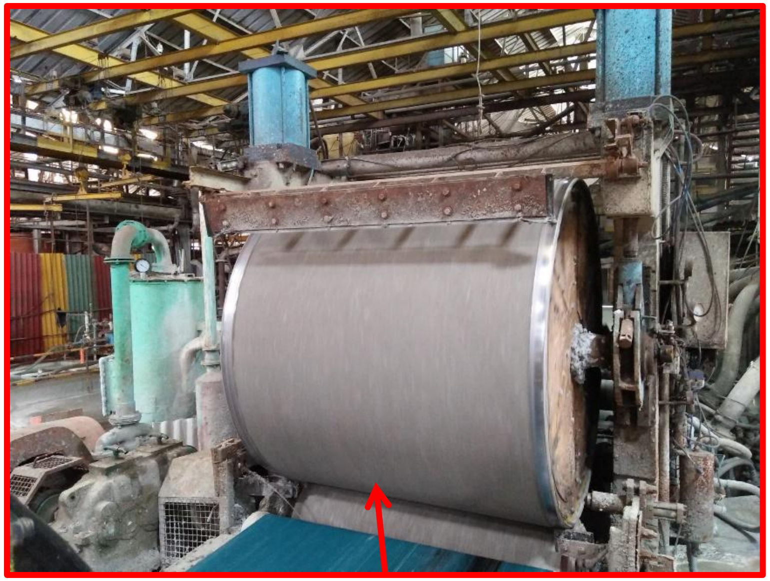
Sheet Collection point

Action: To collect sheets for moulding during cutter change to eliminate machine stoppages & to use pneumatic wrench for changing cutters.

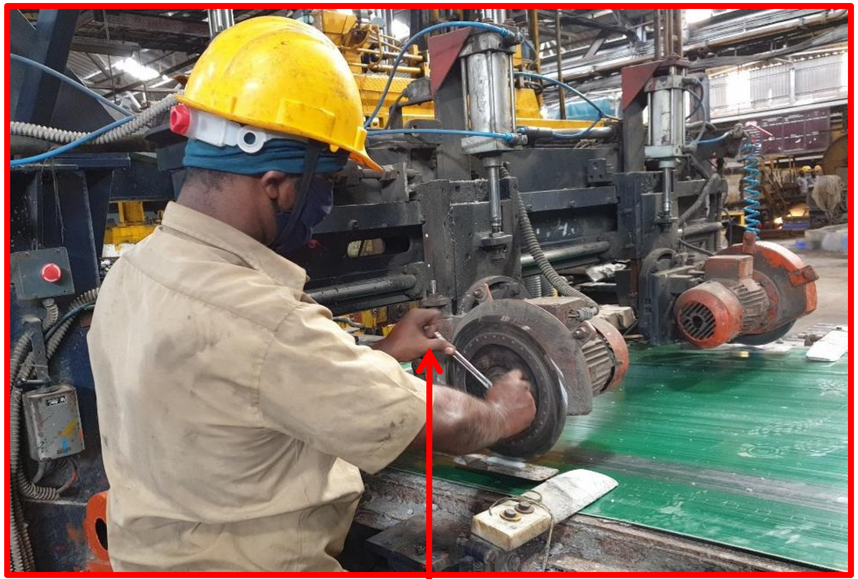
Countermeasure

SN	Counter Measure Activities	Start Date	Target Date	Responsibility	Status
01	To provide individual solenoid valves for each cutter Pneumatic cylinder Up Down movement.	25/10/2019	04/11/2019	Maintenance & Production Department	Completed
02	To procure & use Pneumatic wrench for Screwing & Unscrewing of bolts used for fixing cutters.	25/10/2019	04/11/2019	Production Line Incharge	Completed
03	To collect wet green sheets for Hand Moulding goods while changing cutters.	04/11/2019	04/11/2019	Production Department	Completed
04	To install Conveyor Emergency stop button near cutter unit.	02/11/2019	02/11/2019	Electrical Department	Completed

Before Situation



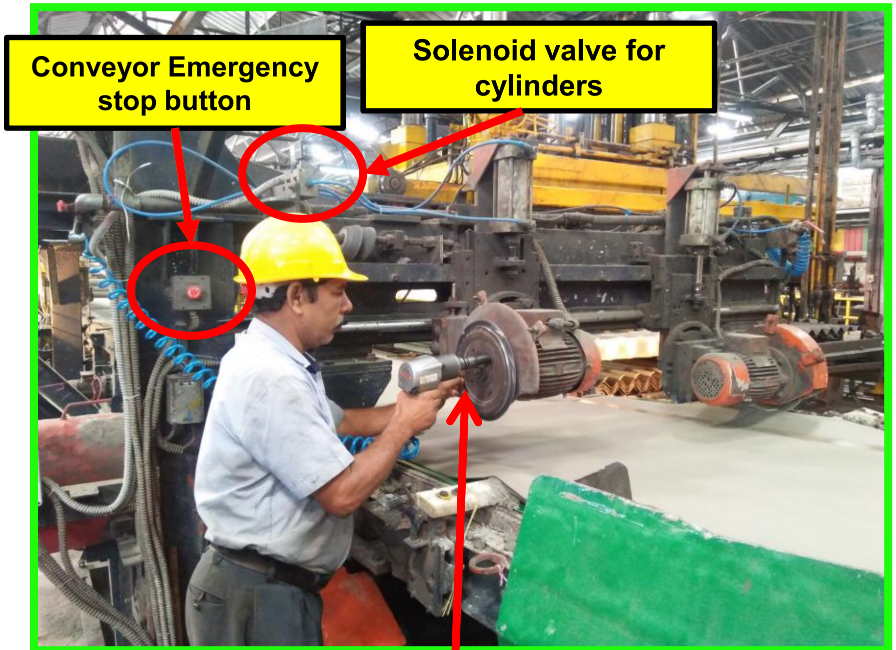
Machine/Bole is stopped



Fitter Changing Cutters with hand spanner

1) For changing cutters we have to stop the machine. If machine is not stopped, green sheets are produced continuosly & we loss them to recycling unit. To avoid loss of green sheets for recycling we are stopping the machine.
2) And cutters are changing with the help of hand spanners. It is time taking process.

After Situation



Fitter Changing Cutters with Pneumatic Wrench

Sheets Collecting for Moulding

- Sheet collecting for moulding goods department & cutter changes has been planned simultaneously. By this, during cutters change, green sheets from machine are collected by moulding department. And there is no Machine stoppage & green sheet loss during cutters change.
- By this, Cutters change time has reduced from 15 minutes to 5 minutes & there is NO machine down time while changing cutters.

After Situation

- Sheet collection for moulding goods department & cutter changes has been planned simultaneously. By this, during cutters change, green sheets from machine are collected by moulding department. And there is no Machine stoppage & green sheet loss during cutters change.
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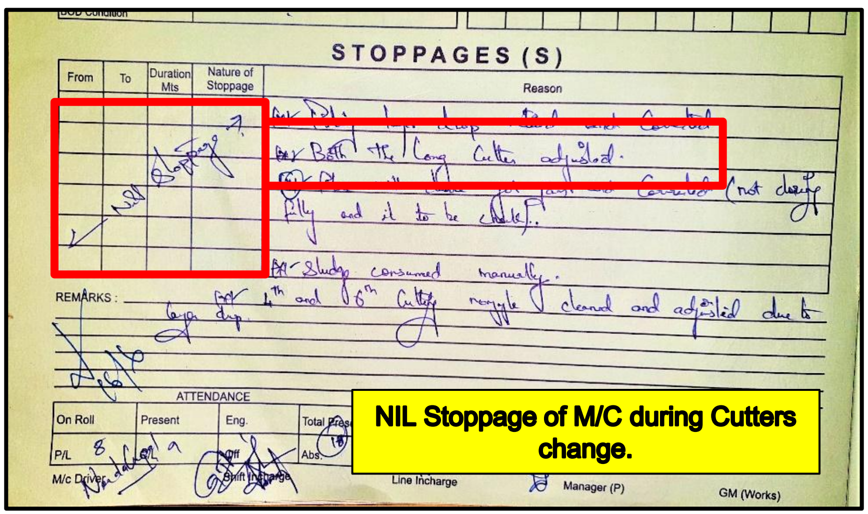
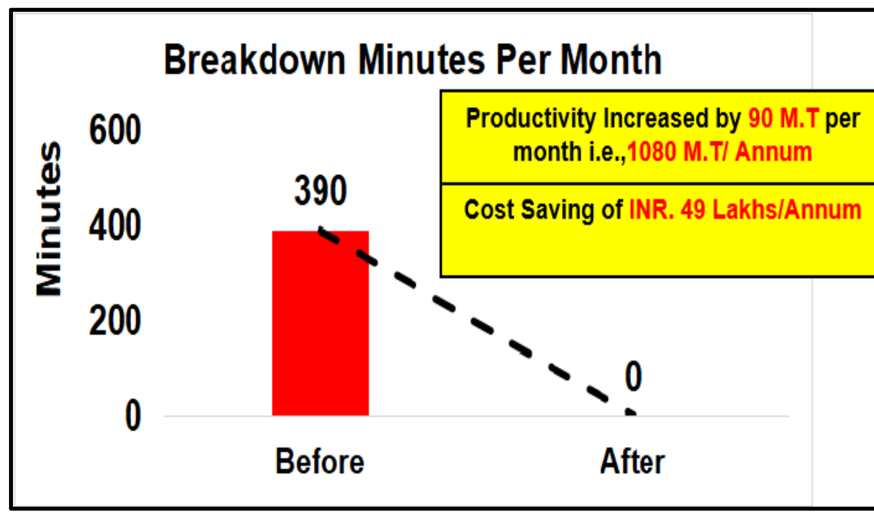
Validation of Data

Increase in Productivity	
After Situation	
Sheet Formation Time	16 Sec
Machine Downtime for changing cutters	15 minutes
Green Sheets produced in 15 minutes	56 sheets
Machine Running days in month	26 Days
Productivity increase interms of M.T/Day	3.52 MT/Day
Productivity increase in terms of M.T/Month	91 MT/Month
Productivity increase in terms of M.T/Year	1098 MT/Year





Benefits Achieved

Qualitative: Cutters are changed without stopping M/C & without losing production. By this M/C downtime for cutters change is eliminated.

Quantitative:

Productivity : Productivity has increased by **90 MT/ Month.**

Cost : Cost Saving of **49 Lakhs/Annum.**

Delivery : Machine capacity has increased by **1098 MT/Year.**

Safety : System is safe with all safety precautions implemented.

Morale : Morale of workmen increased due to reduction in effort for changing cutters & increase in production.

Standardizaiton & Horizontal Deployment

- Training on Updated Standard Operating Procedures are given to all the workmen for regular inspection, maintenance & to work safely.

everest	EVEREST INDUSTRIES LIMITED PODANUR IS/ISO 9001 : 2015 STANDARD OPERATING PROCEDURE	DOC No : EIL/SOP/28 REV. No. : 00 PAGE : 1 OF 1 DATE : 05/11/2019

Standard Operating Procedure: Cutters Change During Machine Running

1) Isolate the power supply & switch OFF the cutter unit.
2) Start giving production to moulding accessories.
3) Lift up the cutter unit by using individual solenoid valve.
4) Change the cutter by using pneumatic gun.
5) Ensure the cutter unit work is completed or not.
6) After completion of cutter change, switch ON the cutter unit.
7) Place cutters in down position with solenoid valve arrangement.
8) Start production of corrugated roofing.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection, Maintenance & safe working.
HOW TO DO	Regular Training on Standard operating procedure.
FREQUENCY	Daily

Horizontal Deployment:

BW	CW	KW	LW	PW1	PW2	SW
Under Progress	Under Progress	Under Progress	Under Progress	Deployed	Deployed	Under Progress

- Under Progress - Deployed

Kaizen Sheet

Category: Reactive

Company	MM/YY	S.No
everest	11/19	02

Reactive ✓	Proactive	Innovative

Kaizen Title: To eliminate Machine stoppages while Cutters & Shoe change.

Implemented Area:
PW1 Shop Floor

Implemented by:
Y. Kesavaraj, Moorthy, Karthikeyan, Madhan Raj, Tulasi Rao, Jagir, Jitenda

Problem/Present Status:

Loss of Production due to Machine stoppages during Cutters and Cutter Shoe changeover.

Before Improvement:

Fitter Changing Cutters with spanner by stopping Machine.

M/C Stopped for changing cutters

Root Cause Identification:

- Problem: Loss of Production
- Why 1: Due to Machine Stoppages.
- Why 2: Machine Stoppages for Cutters & shoes change.
- Why 3: If Machine not stopped for Cutters & shoes change, we loss wet sheet production for recycling.
- Why 4: There is No process to collect Wet sheets during Cutters change without stopping machine.

After Improvement:

Fitter Changing Cutters with Pneumatic Gun

Sheets Collecting for Moulding

Result/Benefit:

Productivity: Increase in productivity of 90 M.T/Month

Cost : cost saving of 49.68 Lakhs/Annum.

Delivery: Cutters Change time has reduced from 15 min to 5 min.

Safety : System is safe.

Morale: By increase in production workmen morale has increased.

Root cause

There is no process to collect green sheets for moulding during cutters change.

Idea to eliminate root cause

To collect the green sheets for moldings department during Cutter change to avoid green sheet loss. Reduce the cutter and cutter shoe changeover time.

Standardization:

Training on SOP & Preventive maitenance as per schedule.

Counter-measure

During sheet collection for moulding, we are making each cutter ideal one by one & simultaneously changing cutters accordingly. And using pneumatic wrench for changing cutters quickly.

Horizontal Deployment

BW	CW	KW	LW	PW	SW